

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
4.	(a)			Indicative content: Treat as neutral reference to cost. Coal Coal is non-renewable. It is reliable and produces large amounts of electricity. Coal is a fossil fuel which produces CO₂ and SO₂ when burned. CO₂ is a greenhouse gas which contributes to climate change. SO₂ produces acid rain. Mining coal damages habitats and the transport of coal also has significant environmental impact. Nuclear Nuclear is non-renewable. It is reliable and it produces very large amounts of electricity and no greenhouse gases. Nuclear produces nuclear waste which is highly radioactive for very long periods of time making it costly and difficult to store. Wind Wind is renewable and it does not produce CO₂, there is also no fuel cost. It is unreliable and to produce sufficient electricity would require very large numbers of turbines. It causes visual and noise pollution. 5–6 marks Discusses advantages and disadvantages of all 3 types of power station. <i>There is a sustained line of reasoning which is coherent, relevant, substantiated and logically structured. The candidate uses appropriate scientific terminology and accurate spelling, punctuation and grammar.</i>	6			6		

Question				Marking details	Marks Available					
					AO1	AO2	AO3	Total	Maths	Prac
				3–4 marks Discusses some advantages and disadvantages of 2 of the 3 types of power station or provides a limited treatment of all. <i>There is a line of reasoning which is partially coherent, largely relevant, supported by some evidence and with some structure. The candidate uses mainly appropriate scientific terminology and some accurate spelling, punctuation and grammar.</i> 1–2 marks Discusses some advantages and disadvantages of 1 of the 3 types of power station or provides a limited treatment of 1 or 2 types. <i>There is a basic line of reasoning which is not coherent, largely irrelevant, supported by limited evidence and with very little structure. The candidate uses limited scientific terminology and inaccuracies in spelling, punctuation and grammar.</i> 0 marks <i>No attempt made or no response worthy of credit.</i>						
	(b)	(i)		10.0 [MWh]		1		1		
		(ii)		Selection and substitution: $\frac{5}{15} [\times 100]$ (1) % efficiency = 33.3 or 33 (1) Award 1 mark for answer only of 0.33		2		2	2	
				Question 4 total	6	3	0	9	2	0